Yik Chun John Peng

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Education

University of Southern California — Los Angeles, CA Expected May 2028
Bachelor of Science, Electrical and Computer Engineering GPA 3.59/4.0
Relevant Coursework: Embedded Systems, Data Structures and Algorithms, Distributed Systems, Digital Logic and RTL Design, Linear Circuits, Internet of Things (IoT)

Skills

Languages: Python, C/C++, Java, Kotlin, MATLAB, Verilog, SQL
Embedded & Hardware: Nordic nRF (nRF52840), ESP32, Raspberry Pi, NVIDIA Jetson, Altium, KiCad, PCBA, Board bring-up, Oscilloscopes
Protocols: CAN, BLE, ESP-NOW, UART, I2C, SPI
Robotics & Vision: ROS 2, Gazebo, MoveIt, URDF, OpenCV, YOLO, OptiTrack motion capture
AI / ML: LangChain, LLM APIs (Gemini, GPT), RAG / vector databases, AI agents, Knowledge graphs (KuzuDB, Neo4j)
Tools: Git/GitHub, Docker, Linux shell, SolidWorks

Experience

Advanced Development Engineering Intern — SRAM, Chicago, IL Jun 2026 – Aug 2026
- Designed a compact Altium PCB for an nRF52840 wireless controller switching an AC interface, balancing electrical performance, cost, manufacturability, and supply-chain constraints.
- Developed nRF52840 firmware in embedded C to drive two independent normally-open relays, reproducing all three interface states with an interlock that prevents both relays from closing and routes every state change through the disconnected state.
- Root-caused a silent Android AIDL/Binder transaction-ID mismatch between a client library and the Karoo service by diffing generated dispatch tables; led to a fixed library release.
- Built internal Monitor Message tooling for the Hammerhead Karoo in Kotlin/Android: eight data fields plus a live graph field that let firmware engineers view any int16 value from a SRAMLink device without adding a new protocol message type.

Machine Learning Engineer Intern, LLM/RAG — Manycore Tech, Hangzhou, China May 2025 – Jul 2025
- Built a knowledge-graph-augmented RAG pipeline using Alibaba Bailian vector retrieval and KuzuDB, improving retrieval relevance by 40% and reducing latency by 20% through optimized graph queries and vector-database integration.
- Developed LangChain agents for KG-driven contextual reasoning and multi-hop retrieval, enabling evidence assembly across graph nodes and semantic search outputs.

Project Manager, Tether — Terra Labs, Los Angeles, CA Sep 2026 – Present
- Architected fully wireless inter-module electronics with one ESP32-class microcontroller per module and no wiring across modules.
- Simplified the robot from 5 modules to 3 self-contained wireless modules by eliminating separate end-effector modules.

Undergraduate Researcher — Robot Locomotion and Navigation Dynamics Laboratory (RoboLAND), USC, Los Angeles, CA Aug 2024 – Present
- Redesigned phase-control algorithms by converting degree-based body-leg and leg-leg coordination into normalized phase offsets, improving gait consistency, interpretability, and cross-condition comparability.
- Executed 46 OptiTrack-tracked physical trials with IR-marker 3D kinematics to characterize performance across multiple coordination regimes.

IoT Systems Intern — MFE Education, Shanghai, China Jun 2024 – Jul 2024
- Developed an IoT device communication library enabling cloud visualization through Huawei Astro Dashboard APIs.

Projects

Cove | Software and Electrical Engineer, Terra Labs Feb 2026 – May 2026
- Designed the control architecture for a 7-DOF robotic arm using ROS 2, Raspberry Pi 5, and CAN-based BLDC actuators.
- Implemented a hardware/software e-stop system with heartbeat timeout, torque cutoff, and ROS 2 fault monitoring.

Circuit Sensei — AI Hardware Agent for Circuit Design and Verification Apr 2026
- Built an AI hardware agent that transforms a natural-language circuit goal into a physically tested Arduino circuit.
- Developed an AI-in-the-loop workflow using Gemma and Gemini Vision to generate step-by-step breadboard diagrams, verify each component placement through a webcam, and block progress until the circuit is assembled correctly.

Vyz — Sensory Adaptation XR Headset | LA TechWeek "Good Vibes Only!" AI/ML Buildathon, Hardware Project Winner Oct 2025
- Designed and built a Jetson-powered sensory-regulation XR headset that detects and mitigates visual and auditory overstimulation in real time using integrated camera, microphone, and embedded-control subsystems.
- Implemented an AI-driven intervention engine that fuses sensor data and triggers GPT-3.5-based parameter selection, achieving under 200 ms latency through shared-state designs optimized for responsiveness and cross-modal adaptation.

SkateMo Autonomous Skateboard | Technical Project Lead / Project Manager, USC Makers Aug 2025 – May 2026
- Led a 6-member multidisciplinary team building an autonomous self-driving skateboard, owning perception, navigation, and embedded-control integration.
- Built real-time obstacle avoidance and path-planning logic using OpenCV and YOLO with camera/IMU fusion; integrated outputs into navigation decisions and safety redundancy.